

SECTION 3: Imperfections in Solids (Q81–110)

81. Schottky defect in NaCl decreases density by:

Slightly, because a small fraction of ions are missing.

✓ A) Slightly

82. Frenkel defect is common in:

Ionic solids with small cations → AgCl, AgBr, ZnS

✓ A) AgCl

83. Colour of NaCl changes due to:

F-centres (electron trapped in anion vacancy)

✓ A) F-centres

84. Vacancy defect increases with:

Temperature (thermal activation)

✓ A) Temperature

85. Fraction of missing atoms = 10^{-6} → density change:

Negligible

✓ A) Negligible

86. Edge dislocation affects:

Mechanical properties (slip & plastic deformation)

✓ A) Mechanical properties

87. Screw dislocation is:

Line defect

✓ A) Line defect

88. F-centres result from:

Electron trapped in anion vacancy

✓ A) Electron trapped in anion vacancy

89. Schottky defect increases with:

Temperature

✓ A) Temperature

90. Frenkel defect increases with:

Temperature

✓ A) Temperature

91. NaCl, fraction = 10^{-6} → missing ions per mole:

$$N_A \times 10^{-6} = 6 \times 10^{17}$$

✓ A) 6×10^{17}

92. Vacancies: $n/N = e^{\{-E/kT\}} = e^{\{-2/(8.617 \times 10^{-5} \times 1000)\}} \approx 10^{-8}$

✓ B) 10^{-8}

93. Colour centres in KCl → blue colour

✓ A) Blue colour

94. Frenkel defect does not affect:

Density (constant) → mainly affects conductivity & diffusion

✓ **A) Density**

95. Schottky defect affects:

Density and electrical conductivity

✓ **D) Both A & B**

96. Edge dislocation line along x-axis. Burgers vector perpendicular →
x-axis

✓ **A) x-axis**

97. F-centres in ionic crystals responsible for:

Colour

✓ **A) Colour**

98. Fraction of Frenkel defects increases with:

Temperature

✓ **A) Temperature**

99. Grain boundary defect is:

Surface defect

✓ **B) Surface defect**

100. Stacking fault occurs due to:

Misalignment in planes

✓ **A) Misalignment in planes**

101. Number of vacancies in Schottky defect at 1200 K, $E_f = 1.2$ eV:

$n/N = e^{\{-E/kT\}} = e^{\{-1.2/(8.617 \times 10^{-5} \times 1200)\}} \approx 10^{-4}$

✓ **A) 10^{-4} fraction**

102. Edge dislocation increases:

Strength (work hardening)

✓ **A) Strength**

103. Frenkel defect common in:

AgBr

✓ **A) AgBr**

104. Schottky defect common in:

NaCl, KCl

✓ **A) NaCl, KCl**

105. F-centres form when:

Electron trapped in anion vacancy

✓ **A) Electron trapped in anion vacancy**

106. Thermal activation increases:

Defect concentration

✓ **A) Defect concentration**

107. Frenkel defect is also called:

Displacement defect

✓ A) Displacement defect

108. Schottky defect occurs in:

NaCl, KCl

✓ A) NaCl, KCl

109. Fraction of missing atoms = 10^{-5} → effect on density:

Slight decrease

✓ A) Slight decrease

110. Edge dislocation line movement causes:

Plastic deformation

✓ A) Plastic deformation

SECTION 4: Electrical & Magnetic Properties (Q111–130)

111. Conductivity of intrinsic semiconductor increases with:

Temperature (more carriers)

✓ A) Temperature

112. Conductivity of metal decreases with:

Temperature increase (more collisions)

✓ A) Temperature increase

113. N-type semiconductor contains:

Extra electrons

✓ A) Extra electrons

114. P-type semiconductor contains:

Extra holes

✓ A) Extra holes

115. Diamagnetic material:

Weakly repelled by magnetic field

✓ A) Weakly repelled

116. Paramagnetic material:

Weakly attracted

✓ A) Weakly attracted

117. Ferromagnetic materials have:

Strong attraction

✓ A) Strong attraction

118. Curie temperature:

Ferromagnetic → paramagnetic transition

✓ A) Ferromagnetic → paramagnetic transition

119. Superconductors have:

Zero resistance

☒ A) Zero resistance

120. Meissner effect:

Expulsion of magnetic field

☒ A) Expulsion of magnetic field

121. Hall effect used to determine:

Type of charge carriers

☒ A) Type of charge carriers

122. Piezoelectric solids generate:

Voltage on stress

☒ A) Voltage on stress

123. Examples of piezoelectric solids:

Quartz, Rochelle salt

☒ A) Quartz, Rochelle salt

124. Magnetic domains are:

Regions of aligned magnetic moments

☒ A) Regions of aligned magnetic moments

125. Paramagnetic susceptibility:

Positive

☒ A) Positive

126. Diamagnetic susceptibility:

Negative

☒ A) Negative

127. Superconductivity destroyed above:

Critical temperature

☒ A) Critical temperature

128. Conductivity of extrinsic semiconductor increases with:

Doping

☒ A) Doping

129. Example of ferromagnetic material:

Fe

☒ A) Fe

130. Example of diamagnetic material:

Cu

☒ A) Cu